SE Lab 7

Fuzzing for Input Validation Bugs

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Date: 11th Nov, 2026

Screenshots:

SS1:

```
ronak@Ronaks-MacBook-Pro-4 LAB7 % python3 car_insurance_premium_parser.py
==== Running normalize_registration ====
Enter a registration number (e.g. ka 01 ab 1234): ka 21 cm 2831
Normalized registration: KA21CM2831

==== Running total_claim_amount ====
Enter past claim amounts separated by ; (e.g. 12000;4500;800): 8000;4000;500
Total claimed: 12500

==== Running driver_age ====
Enter the driver's date of birth as DD/MM/YYYY (e.g. 14/08/1999): 21/07/1996
Driver age in 2026: 30
ronak@Ronaks-MacBook-Pro-4 LAB7 %
```

SS2:

```
ronak@Ronaks-MacBook-Pro-4 LAB7 % pytest test_car_insurance_premium_parser.py
============================= test session starts ==============================
platform darwin -- Python 3.13.6, pytest-9.1.1, pluggy-1.6.0
rootdir: /Users/ronak/PES_progs/5th_sem_progs/SE_LAB/Lab 7 Fuzzing for Input Validation Bugs/SE_L6_L7_55_car_insurance_premium/LAB7
plugins: hypothesis-6.170.15, Faker-40.21.0, anyio-4.13.0
collected 3 items

test_car_insurance_premium_parser.py FFF                                 [100%]

=================================== FAILURES ===================================
_____________________ test_normalize_registration_no_crash _____________________

s = None

    @given(st.text() | st.none())
    def test_normalize_registration_no_crash(s):
        try:
>           normalize_registration(s)

test_car_insurance_premium_parser.py:7:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

plate = None

    def normalize_registration(plate):
        """
        Removes spaces and dashes from a registration number and upper-cases it.
        Assumes plate is a non-empty string.
        """
>       plate = plate.strip().upper()
                ^^^^^^^^^^^^
E       AttributeError: 'NoneType' object has no attribute 'strip'

car_insurance_premium_parser.py:6: AttributeError

During handling of the above exception, another exception occurred:

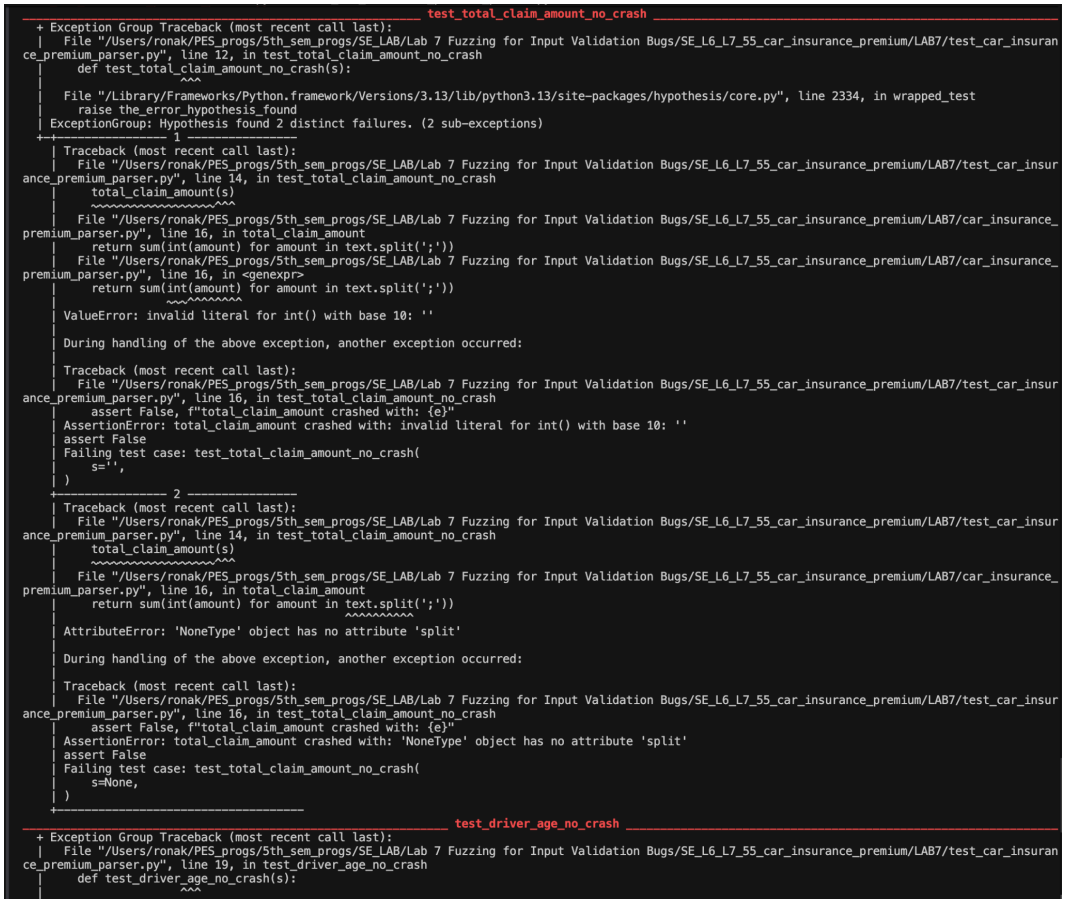
    @given(st.text() | st.none())
>   def test_normalize_registration_no_crash(s):
                ^^^

test_car_insurance_premium_parser.py:5:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

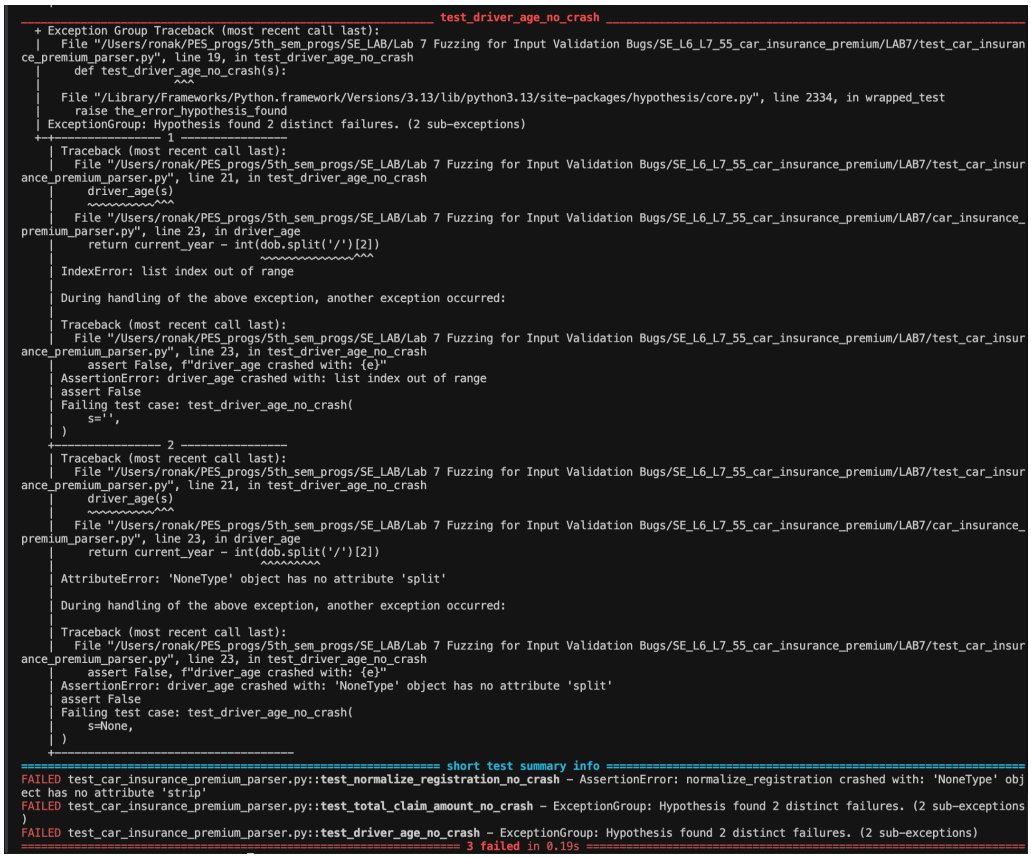
s = None

    @given(st.text() | st.none())
    def test_normalize_registration_no_crash(s):
        try:
            normalize_registration(s)
        except Exception as e:
>           assert False, f"normalize_registration crashed with: {e}"
E           AssertionError: normalize_registration crashed with: 'NoneType' object has no attribute 'strip'
E           assert False
E           Failing test case: test_normalize_registration_no_crash(
E               s=None,
E           )

test_car_insurance_premium_parser.py:9: AssertionError
```

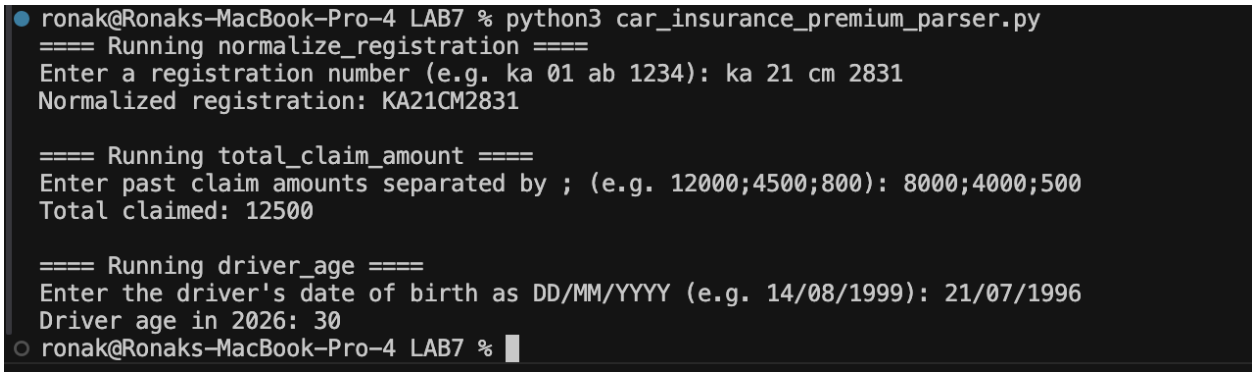


```
_____ test_total_claim_amount_no_crash _____
  + Exception Group Traceback (most recent call last):
  |   File "/Users/ronak/PES_progs/5th_sem_progs/SE_LAB/Lab 7 Fuzzing for Input Validation Bugs/SE_L6_L7_55_car_insurance_premium/LAB7/test_car_insurance_premium_parser.py", line 12, in test_total_claim_amount_no_crash
  |     def test_total_claim_amount_no_crash(s):
  |                ^^^
  |   File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/hypothesis/core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/ronak/PES_progs/5th_sem_progs/SE_LAB/Lab 7 Fuzzing for Input Validation Bugs/SE_L6_L7_55_car_insurance_premium/LAB7/test_car_insurance_premium_parser.py", line 14, in test_total_claim_amount_no_crash
    |     total_claim_amount(s)
    |     ^^^^^^^^^^^^^^^^^^^^^
    |   File "/Users/ronak/PES_progs/5th_sem_progs/SE_LAB/Lab 7 Fuzzing for Input Validation Bugs/SE_L6_L7_55_car_insurance_premium/LAB7/car_insurance_premium_parser.py", line 16, in total_claim_amount
    |     return sum(int(amount) for amount in text.split(';'))
    |   File "/Users/ronak/PES_progs/5th_sem_progs/SE_LAB/Lab 7 Fuzzing for Input Validation Bugs/SE_L6_L7_55_car_insurance_premium/LAB7/car_insurance_premium_parser.py", line 16, in <genexpr>
    |     return sum(int(amount) for amount in text.split(';'))
    |                ^^^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/ronak/PES_progs/5th_sem_progs/SE_LAB/Lab 7 Fuzzing for Input Validation Bugs/SE_L6_L7_55_car_insurance_premium/LAB7/test_car_insurance_premium_parser.py", line 16, in test_total_claim_amount_no_crash
    |     assert False, f"total_claim_amount crashed with: {e}"
    | AssertionError: total_claim_amount crashed with: invalid literal for int() with base 10: ''
    | assert False
    | Failing test case: test_total_claim_amount_no_crash(
    |     s='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/ronak/PES_progs/5th_sem_progs/SE_LAB/Lab 7 Fuzzing for Input Validation Bugs/SE_L6_L7_55_car_insurance_premium/LAB7/test_car_insurance_premium_parser.py", line 14, in test_total_claim_amount_no_crash
    |     total_claim_amount(s)
    |     ^^^^^^^^^^^^^^^^^^^^^
    |   File "/Users/ronak/PES_progs/5th_sem_progs/SE_LAB/Lab 7 Fuzzing for Input Validation Bugs/SE_L6_L7_55_car_insurance_premium/LAB7/car_insurance_premium_parser.py", line 16, in total_claim_amount
    |     return sum(int(amount) for amount in text.split(';'))
    |                                           ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/ronak/PES_progs/5th_sem_progs/SE_LAB/Lab 7 Fuzzing for Input Validation Bugs/SE_L6_L7_55_car_insurance_premium/LAB7/test_car_insurance_premium_parser.py", line 16, in test_total_claim_amount_no_crash
    |     assert False, f"total_claim_amount crashed with: {e}"
    | AssertionError: total_claim_amount crashed with: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_total_claim_amount_no_crash(
    |     s=None,
    | )
    +------------------------------------
_____ test_driver_age_no_crash _____
  + Exception Group Traceback (most recent call last):
  |   File "/Users/ronak/PES_progs/5th_sem_progs/SE_LAB/Lab 7 Fuzzing for Input Validation Bugs/SE_L6_L7_55_car_insurance_premium/LAB7/test_car_insurance_premium_parser.py", line 19, in test_driver_age_no_crash
  |     def test_driver_age_no_crash(s):
  |                ^^^
  |
```

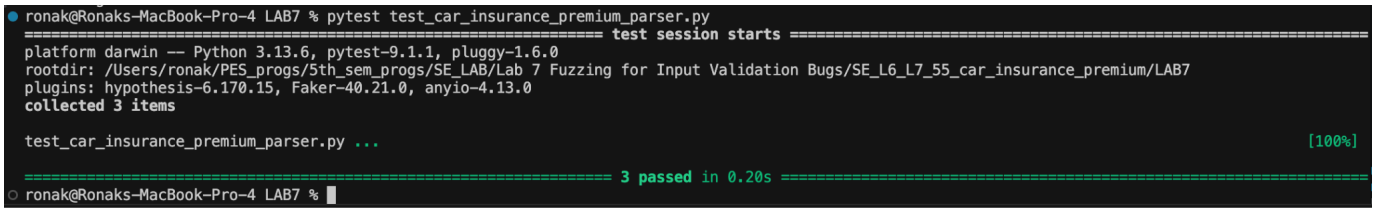


```
_____ test_driver_age_no_crash _____
  + Exception Group Traceback (most recent call last):
  |   File "/Users/ronak/PES_progs/5th_sem_progs/SE_LAB/Lab 7 Fuzzing for Input Validation Bugs/SE_L6_L7_55_car_insurance_premium/LAB7/test_car_insurance_premium_parser.py", line 19, in test_driver_age_no_crash
  |     def test_driver_age_no_crash(s):
  |                ^^^
  |   File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/hypothesis/core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/ronak/PES_progs/5th_sem_progs/SE_LAB/Lab 7 Fuzzing for Input Validation Bugs/SE_L6_L7_55_car_insurance_premium/LAB7/test_car_insurance_premium_parser.py", line 21, in test_driver_age_no_crash
    |     driver_age(s)
    |     ^^^^^^^^^^^^^
    |   File "/Users/ronak/PES_progs/5th_sem_progs/SE_LAB/Lab 7 Fuzzing for Input Validation Bugs/SE_L6_L7_55_car_insurance_premium/LAB7/car_insurance_premium_parser.py", line 23, in driver_age
    |     return current_year - int(dob.split('/')[2])
    |                           ^^^^^^^^^^^^^^^^^^^^^^
    | IndexError: list index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/ronak/PES_progs/5th_sem_progs/SE_LAB/Lab 7 Fuzzing for Input Validation Bugs/SE_L6_L7_55_car_insurance_premium/LAB7/test_car_insurance_premium_parser.py", line 23, in test_driver_age_no_crash
    |     assert False, f"driver_age crashed with: {e}"
    | AssertionError: driver_age crashed with: list index out of range
    | assert False
    | Failing test case: test_driver_age_no_crash(
    |     s='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/ronak/PES_progs/5th_sem_progs/SE_LAB/Lab 7 Fuzzing for Input Validation Bugs/SE_L6_L7_55_car_insurance_premium/LAB7/test_car_insurance_premium_parser.py", line 21, in test_driver_age_no_crash
    |     driver_age(s)
    |     ^^^^^^^^^^^^^
    |   File "/Users/ronak/PES_progs/5th_sem_progs/SE_LAB/Lab 7 Fuzzing for Input Validation Bugs/SE_L6_L7_55_car_insurance_premium/LAB7/car_insurance_premium_parser.py", line 23, in driver_age
    |     return current_year - int(dob.split('/')[2])
    |                           ^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/ronak/PES_progs/5th_sem_progs/SE_LAB/Lab 7 Fuzzing for Input Validation Bugs/SE_L6_L7_55_car_insurance_premium/LAB7/test_car_insurance_premium_parser.py", line 23, in test_driver_age_no_crash
    |     assert False, f"driver_age crashed with: {e}"
    | AssertionError: driver_age crashed with: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_driver_age_no_crash(
    |     s=None,
    | )
    +------------------------------------
=========================== short test summary info ============================
FAILED test_car_insurance_premium_parser.py::test_normalize_registration_no_crash - AssertionError: normalize_registration crashed with: 'NoneType' object has no attribute 'strip'
FAILED test_car_insurance_premium_parser.py::test_total_claim_amount_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_car_insurance_premium_parser.py::test_driver_age_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
============================== 3 failed in 0.19s ===============================
```

SS3:


ronak@Ronaks-MacBook-Pro-4 LAB7 % python3 car_insurance_premium_parser.py
==== Running normalize_registration ====
Enter a registration number (e.g. ka 01 ab 1234): ka 21 cm 2831
Normalized registration: KA21CM2831
==== Running total_claim_amount ====
Enter past claim amounts separated by ; (e.g. 12000;4500;800): 8000;4000;500
Total claimed: 12500
==== Running driver_age ====
Enter the driver's date of birth as DD/MM/YYYY (e.g. 14/08/1999): 21/07/1996
Driver age in 2026: 30
ronak@Ronaks-MacBook-Pro-4 LAB7 %

SS4:


ronak@Ronaks-MacBook-Pro-4 LAB7 % pytest test_car_insurance_premium_parser.py
test session starts
platform darwin -- Python 3.13.6, pytest-9.1.1, pluggy-1.6.0
rootdir: /Users/ronak/PES_progs/5th_sem_progs/SE_LAB/Lab 7 Fuzzing for Input Validation Bugs/SE_L6_L7_55_car_insurance_premium/LAB7
plugins: hypothesis-6.170.15, Faker-40.21.0, anyio-4.13.0
collected 3 items
test_car_insurance_premium_parser.py ...
[100%]
3 passed in 0.20s
ronak@Ronaks-MacBook-Pro-4 LAB7 %